display_image_opencv

This sample application is designed to test an OpenCV algorithm applied to a JPEG image. As it is built upon another sample application called 'display_image', it allows testing of the Hanwha Vision Open Platform's Network Manager. By using the Network Manager, an image processed with the OpenCV Canny algorithm can be displayed on the application's web page. For more details, refer to **3.1 Network Manager** in **Hanwha_Vision_OpenPlatform_24.00.01_SDK_API**.

Related events are as follows:

Sendable Events

Event type	Header file	Event type (enum)
Start Service	i_app_network_manager.h	IAppNetworkManager::EEventType::eStartService
Stop Service	i_app_network_manager.h	IAppNetworkManager::EEventType::eStopService
Send Data	i_app_network_manager.h	IAppNetworkManager::EEventType::eSendData
Close Client	i_app_network_manager.h	IAppNetworkManager::EEventType::eCloseClient

Receivable Events

Event type	Header file	Event type (enum)
New Client Connected	i_app_network_manager.h	IAppNetworkManager::EEventType::eNewClientConnected
Client Disconnected	i_app_network_manager.h	IAppNetworkManager::EEventType::eClientDisconnected
Client Data	i_app_network_manager.h	IAppNetworkManager::EEventType::eClientData
Server Data	i_app_network_manager.h	IAppNetworkManager::EEventType::eServerData

Prerequisites

- Hanwha Vision Open Platform 23.00.00 or higher

Scenario

- Preconditions
  1. Open the camera web viewer, go to [Setup]>[Open Platform], and click [Install] to install the application.

2. Click [Start] to start the application.

- Test
  1. In the [Open Platform] menu of the web viewer, click [Go App].
  2. Enter port number in checkbox and click [Start Server] on the application web page.
  3. Click [Request] on the application web page.

Result

- A "Pass" message will appear below the [Start Server] button.
- An image will be displayed under [3. Display Image].

1. Start Server

port 8080

Start Server

Pass

2. Request Image

Request

3. Display Image

Building Application (cv5)

1. Download and cross compile OpenCV source code.

```
opencv$ mkdir build
opencv$ cd build
```

```
opencv/build$ cmake -DCMAKE_TOOLCHAIN_FILE=../platforms/linux/aarch64
-gnu.toolchain.cmake ..
opencv/build$ make install
```

2. Copy libraries and headers. Let’s say path of opencv on step 1 above is [opencv]

```
display_image_opencv/app/libs$ cp -a [opencv]/build/install/lib/* .
display_image_opencv/app/libs$ cd ../includes
display_image_opencv/app/includes$ cp -a
[opencv]/build/install/include/* .
```

3. Build application.

```
$ APP_NAME=display_image_opencv SDK_VER=24.06.14(your SDK version)
SOC=[cv5, jetson] docker compose up
$ docker compose down --remove-orphans
```

4. Check the build results in current directory. If successful, you will be able to find the cap file.